

# Possible topological relationships between spatial objects -1

## Point-Point Relationship

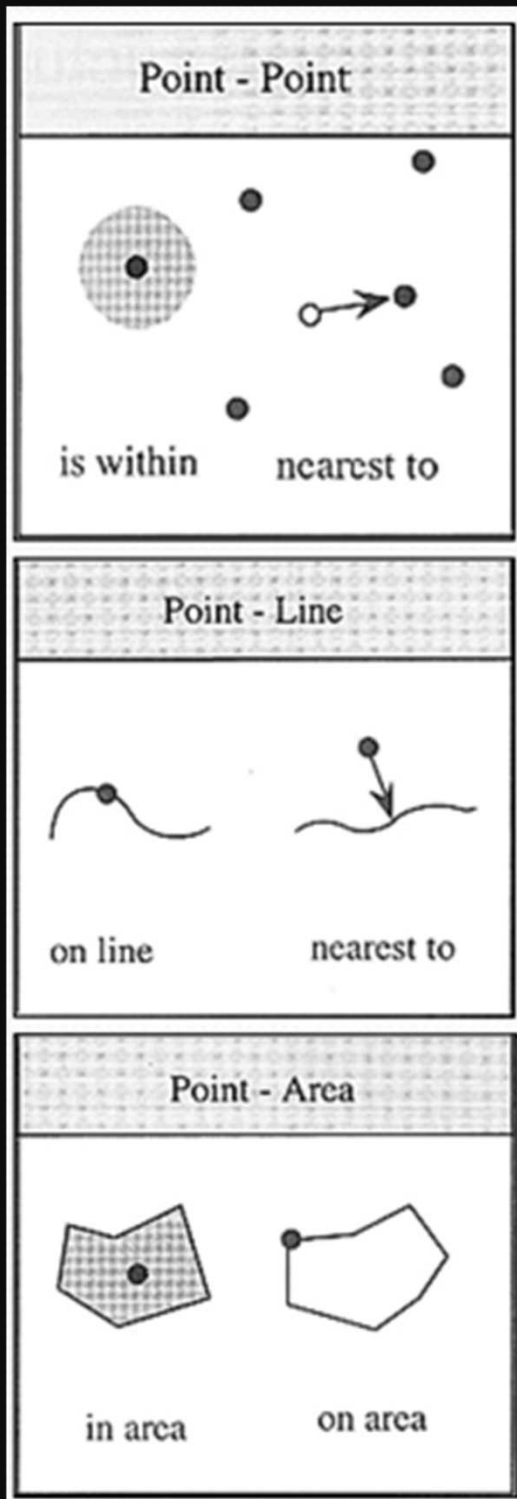
- "is within" : within a certain distance
- "is nearest to" : nearest to a certain point

## Point-Line Relationships

- "on line" : a point on a line
- "is nearest to" : a point nearest to a line

## Point-area Relationships

- "is contained in" : a point in an area
- "on border of area" : a point on border of an area



# Possible topological relationships between spatial objects-2

## Line-Line Relationships

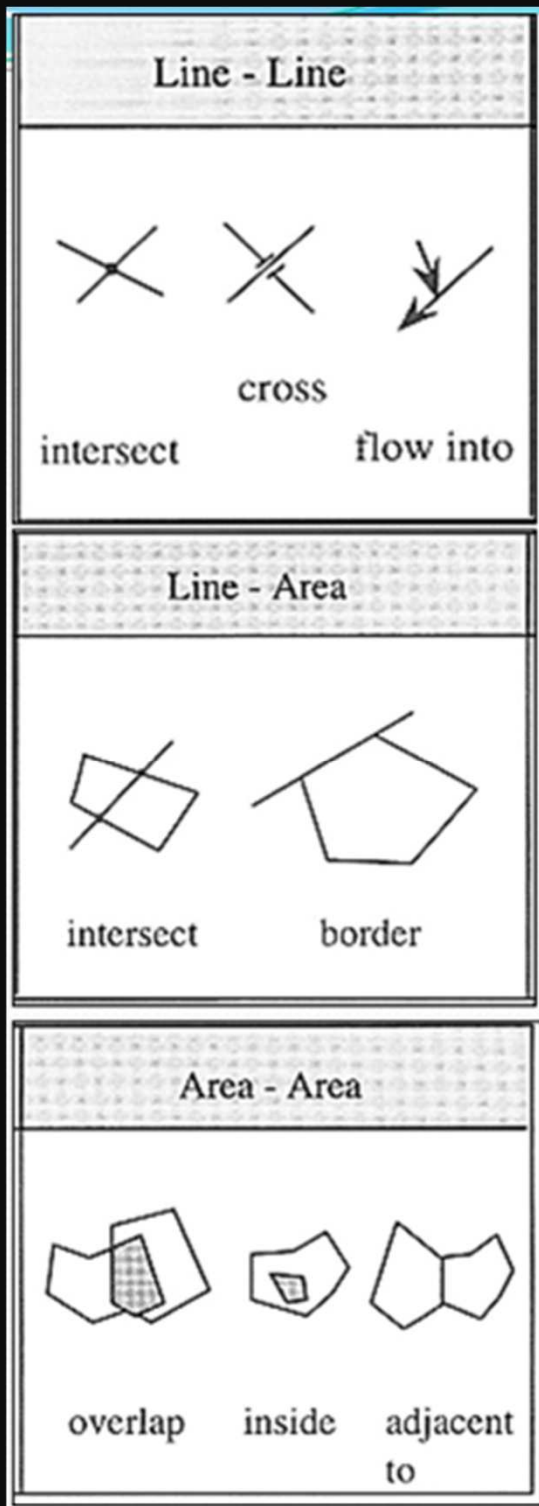
- "intersects" : two lines intersect
- "crosses" : two lines cross without an intersect
- "flow into" : a stream flows into the river

## Line-Area Relationship

- "intersects" : a line intersects an area
- "borders" : a line is a part of border of an area

## Area-Area Relationships

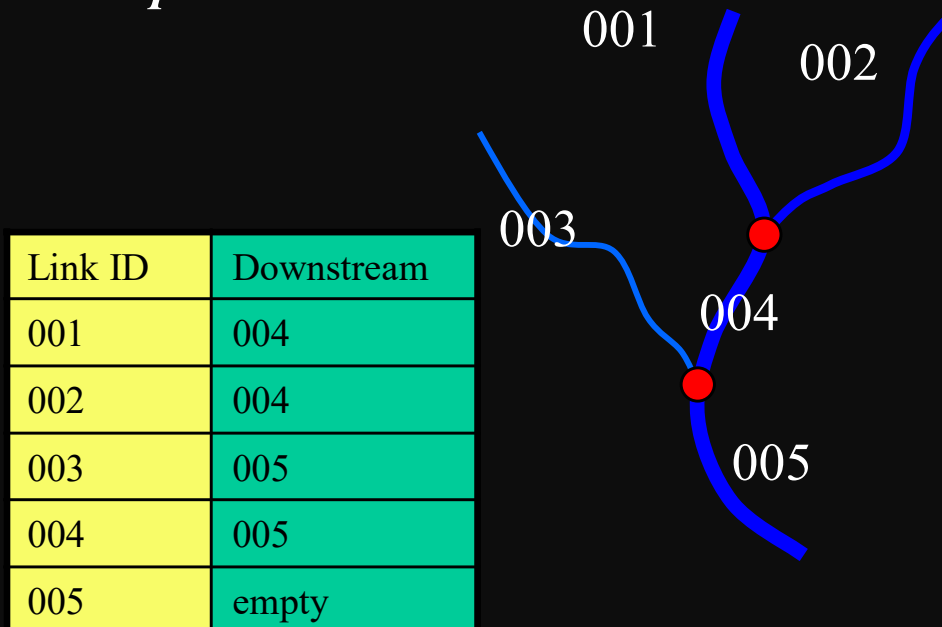
- "overlaps" : two areas overlap
- "is within" : an island within an area
- "is adjacent to" : two area share a common boundary



# Relationships as attributes

- ◆ Example: “Flows-in” relationship

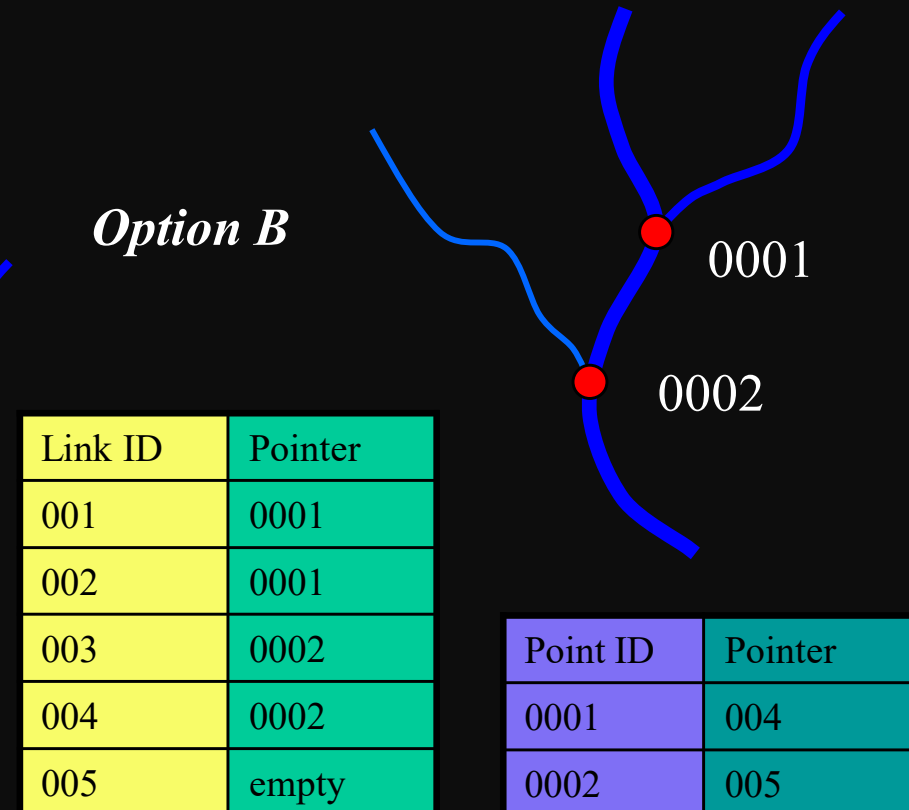
*Option A*



Each stream link in a stream network could be given the ID of the downstream link which it flows into

Flow could be traced from link to link by following pointers

*Option B*



Each stream link in a stream network could be given the ID of the downstream point which it flows into

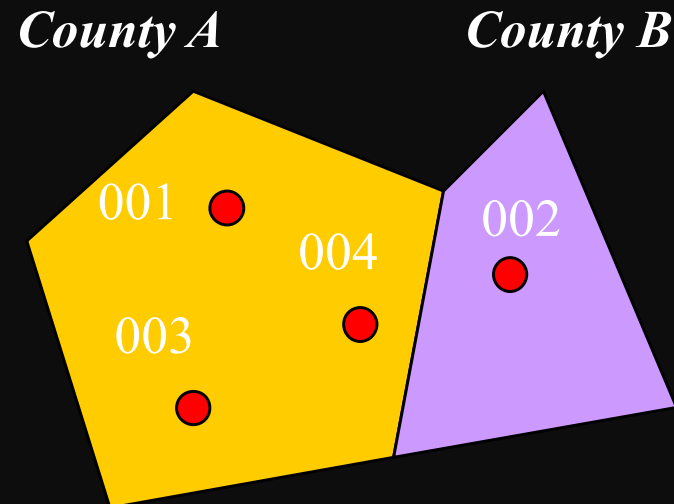
Each stream point in a stream network could be given the ID of the downstream link which it flows into

Flow could be traced from link to link by following pointers

# Relationships as attributes

◆ Example: “is contained in” relationship

1. Find the containing county of each well (compute the “is contained in” relationship).
2. Store the result as a new attribute, **County**, of each well.
3. Using this revised attribute table, total flow by county and add results to the county table.



| Well ID | County |
|---------|--------|
| 001     | A      |
| 002     | B      |
| 003     | A      |
| 004     | A      |



| County ID | No. of Wells |
|-----------|--------------|
| A         | 3            |
| B         | 1            |

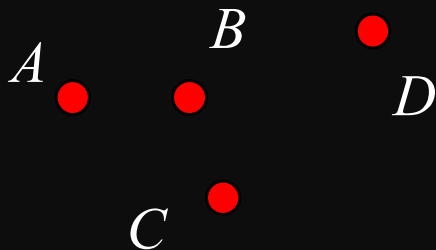
# Object Pairs

- ◆ Some attribute is between a pair of objects.
  - ◆ Distance is the attribute of a pair of objects.
  - ◆ Flow of commuters between two places.
  - ◆ Trade between two counties

In some cases these attributes can be attached to an object linking the origin and destination objects.

For example, trade can be attached to an arrow pointing from county A to B.

- ◆ In general, these kind of attributes shall be described using separate tables or matrix.



|   | A   | B | C   | D   |
|---|-----|---|-----|-----|
| A | 0   | 1 | 1.5 | 2.5 |
| B | 1   | 0 | 1   | 2   |
| C | 1.5 | 1 | 0   | 3   |
| D | 2.5 | 2 | 2.5 | 0   |

*A distance table between points A, B, C, and D*

# Topology

- “A GIS topology is a *set of rules* and *behaviors* that *model how points, lines, and polygons share Geometry*” *ESRI*
- “Topology is the science and mathematics of *relationships* used to *validate the geometry* of *vector* entities, and for operations such as *network* tracing and tests of polygon *adjacency*” *Longley*

*Topology is a mathematics approach that defines unchangeable spatial relationships.*

# Spatial relationships and Topology

- ◆ When a map is stretched or distorted, some properties change,
  - ◆ Distance ; Angles; Relative proximities
- ◆ Some properties won't change,
  - ◆ Adjacencies
  - ◆ Most other relationships, such as "is contained in", "crosses"
  - ◆ Types of spatial objects - areas remain areas, lines remain lines, points remain points
- ◆ These unchanged properties are called topological properties.

## Advantages of Topology

1. Efficient data storage, Quick Processing of large data set
2. Spatial model operations

# Types of Topological Relationships

## **1) Connectivity**

- Indicates which geographic features connect to others or which geographic features intersect each other;

## **2) Adjacency**

- Indicates which geographic features are adjacent (contiguous) to others;

## **3) Containment (Nestedness, Insideness)**

- Indicates which geographic features (node, arc, smaller polygon) are contained within a polygon.

## **4) Proximity**

- Indicates which geographic features are near others;

## **5) Relative Direction**

- Indicates the relative position between geographic features;

## Representation of Connectivity by Arc-node Topology

- An arc is defined by two endpoints, the from-node indicating where the arc begins and a to-node indicating where it ends.

This **arc-node topology** indicates which arcs are connected at which nodes;

- Arc-node topology is supported through an arc-node list.  
Connected arcs are determined by searching through the list for **common node numbers**.
- A set of connected arcs can define a network.

## Representation of Direction by from-node and to-node

- The from-node and to-node define an arc and also indicates an arc's direction;
- This makes it possible to represent flow data;

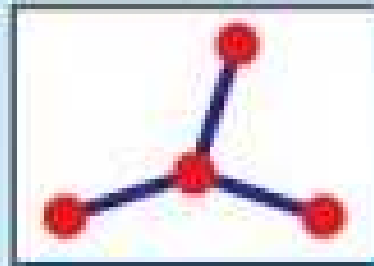
## Representation of Adjacency

- Two geographic features that share a boundary are called adjacent.
- For each arc, the polygons on its left and right can be determined from the arc's direction. This is referred to as left-right topology.

# Topology

Line features can share endpoints

*arc-node topology*



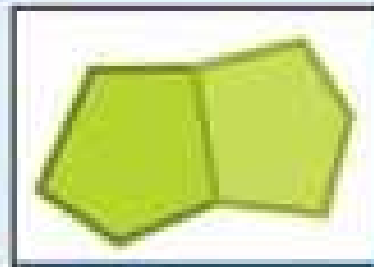
Area features can overlap with other area features

*region topology*



Area features can share boundaries

*polygon topology*



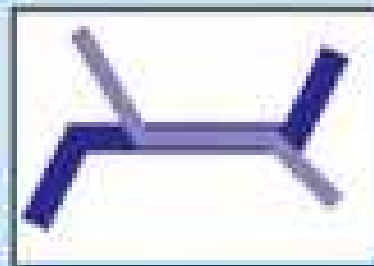
Line features can share endpoint vertices with point features

*node topology*



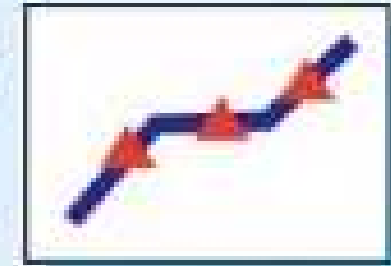
Line features can share segments with other line features

*route topology*



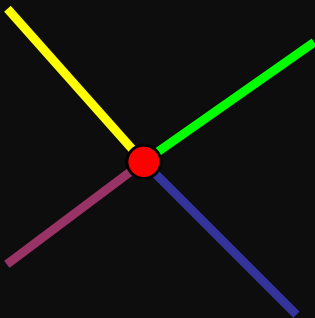
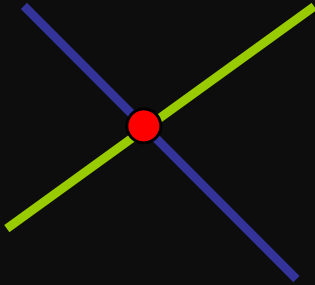
Point features can share vertices with line features

*point events*

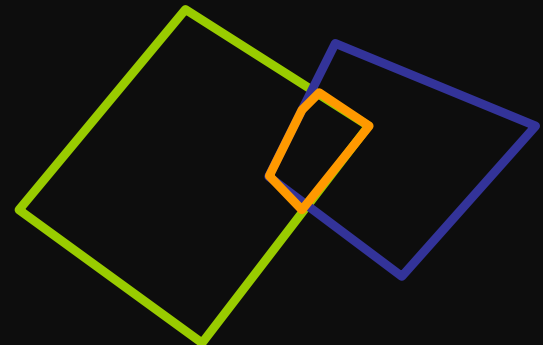
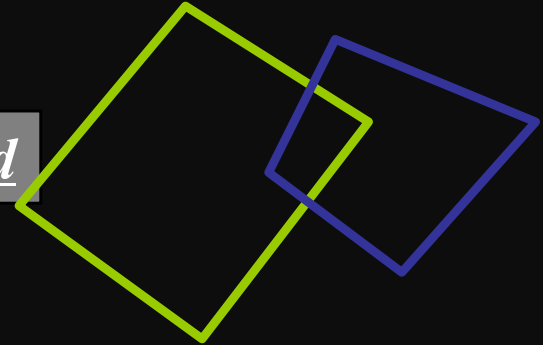


# Topological examples

*Network connectivity*



*Polygon adjacency*



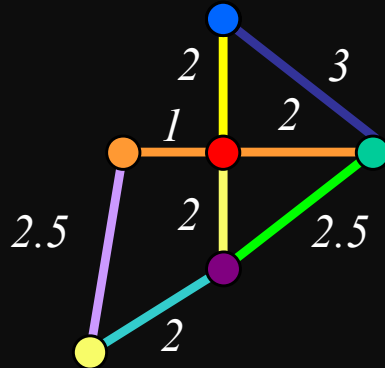
*Topology poorly-defined*

*Topology well-defined*

## Importance of topology

# Topology and GIS analysis

## *Searching a shortest path*

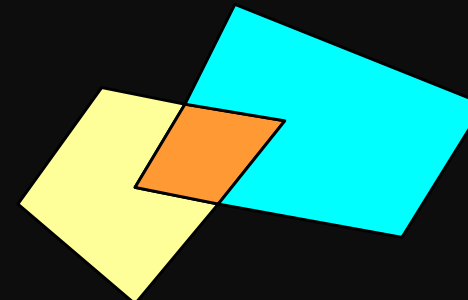


The shortest path from the blue point to the yellow point is through the red point and then the orange point ( $2+1+2.5=5.5$  map units).

However, if the topology of the red point is not defined clearly, which means the two yellow lines are considered as one and the two orange lines are considered as one, the resulting answer will be wrong ( $2+2+2=6$  map units).

The overlapped two polygons have to be cut into three in order to clearly define the spatial topology.

Otherwise there will be difficulties finding an adjacent polygon of either.



*Finding adjacent areas*